

Introduction

Sports injuries and overuse conditions are increasingly common in athletes, which created a growing demand for effective recovery methods. Ice therapy is widely used to reduce pain and inflammation; however, its cooling effects are primarily limited to superficial tissue layers (Bleakley et al.). In contrast, cupping therapy applies suction to lift the skin and underlying tissue, which may enhance blood flow and influence how treatments interact with deeper layers(Khoshnevis et al.). Recent research suggests that many recovery techniques fail to effectively target deep muscle tissue, limiting their overall effectiveness (Dupuy et al.). Because cooling and suction impact tissue differently, combining these methods may improve therapeutic outcomes. This study investigates whether integrating cryotherapy with suction—through a cryogenic cupping system can enhance cooling depth within a simulated human tissue model.

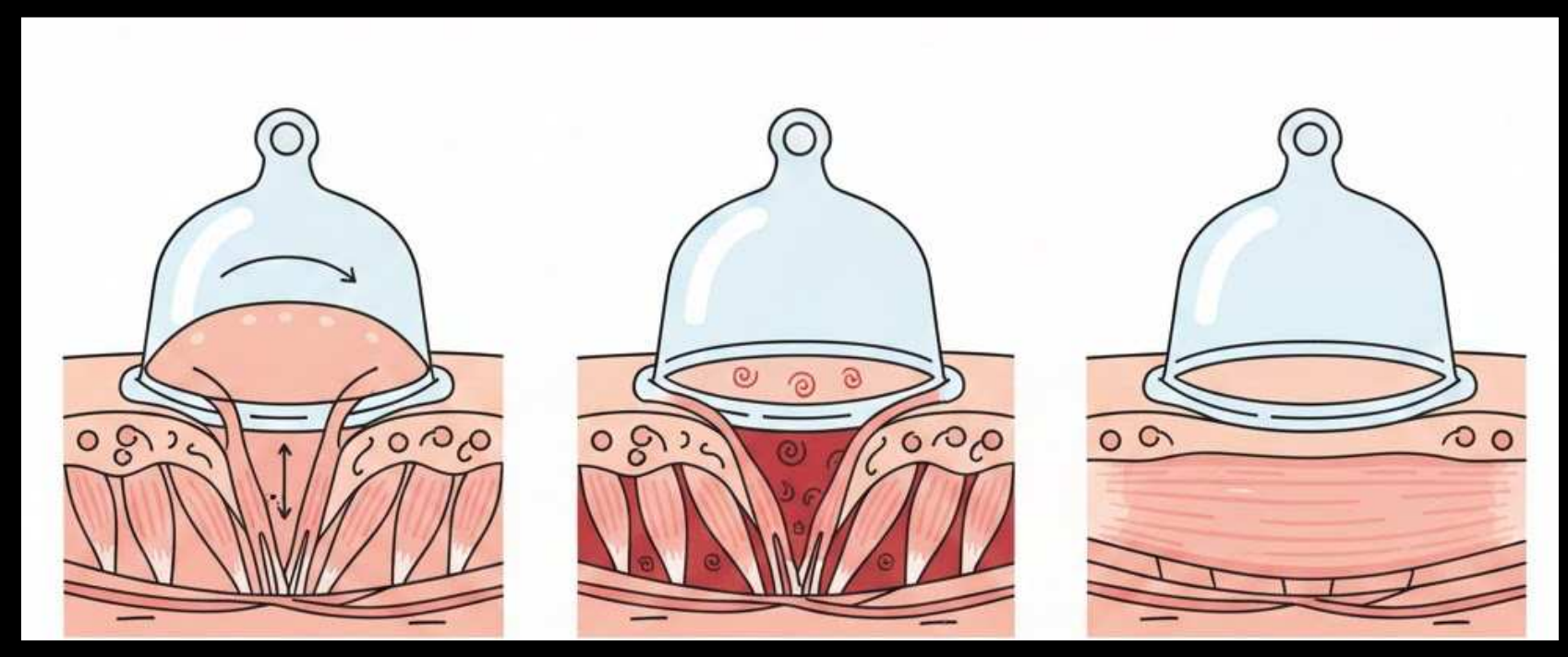


Figure 1. Diagram that shows stages of cupping: lifting skin, drawing blood flow, muscle relaxation

Purpose

Cryogenic cupping was tested to evaluate whether combining suction with cooling could improve cooling depth compared to traditional ice therapy. Temperature changes were measured at the surface, mid-layer, and deep tissue regions of a simulated tissue model to compare how effectively each method transferred cooling through tissue over time.

Cryogenic Cupping and Its Effect on Cooling Depth in Simulated Human Tissue

Methods

Phantom Limb



Figure 2. Layered phantom forearm constructed to simulate human tissue.

Ballistic gel was used to model surface, mid-layer, and deep tissue regions, mimicking how cooling travels through the body. Thermistors were embedded at each depth to measure temperature changes over time. This allowed for direct comparison of how far cooling effects penetrated into the model.

Cryogenic Cups

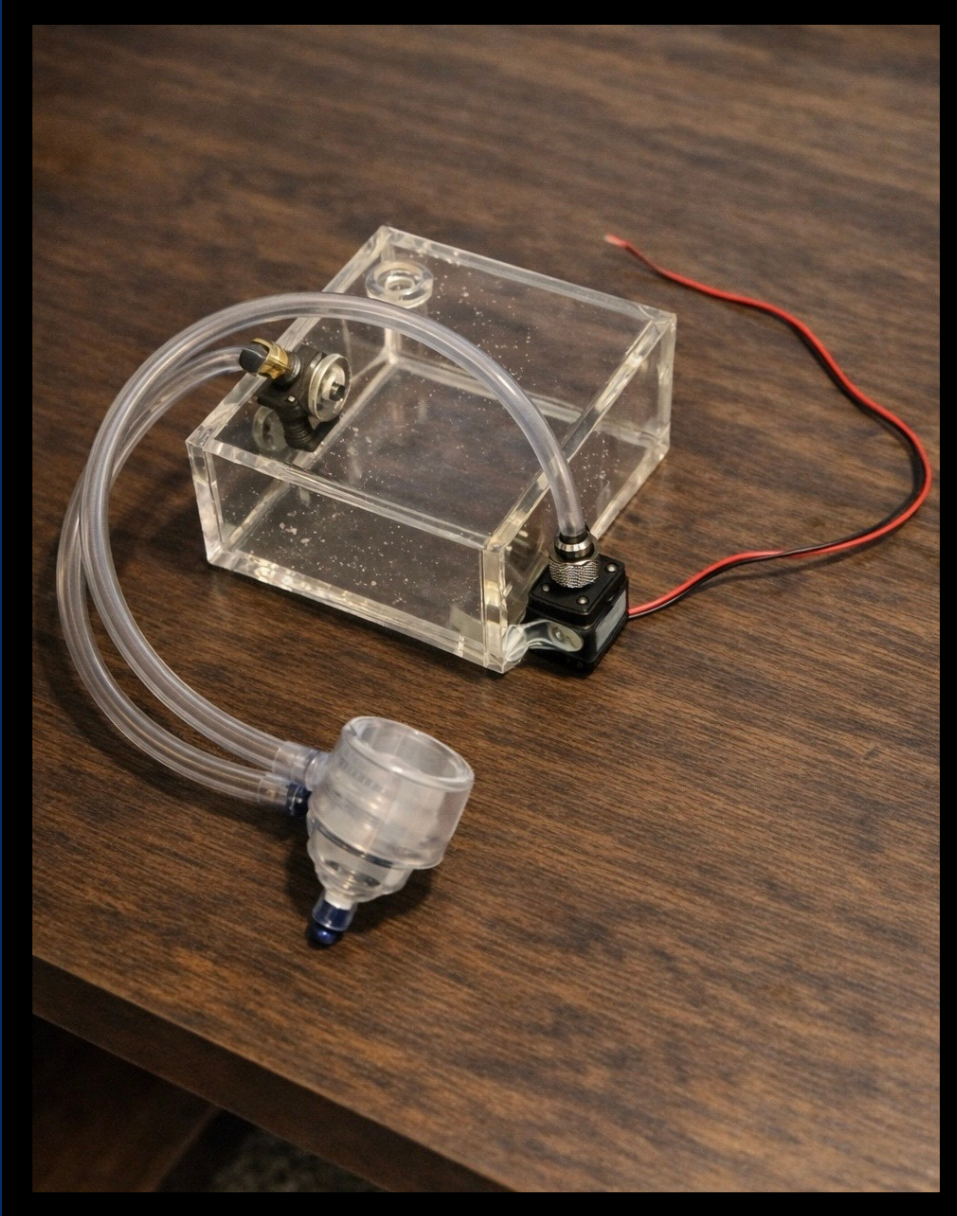


Figure 3. Cryogenic cupping system with circulating cold fluid.

A pump-driven reservoir system circulated cold water through tubing surrounding the suction cup. It has continuous cooling while suction is applied, compared to traditional ice therapy. The system was built to test whether combining these methods improves cooling depth and consistency.

Physical Testing

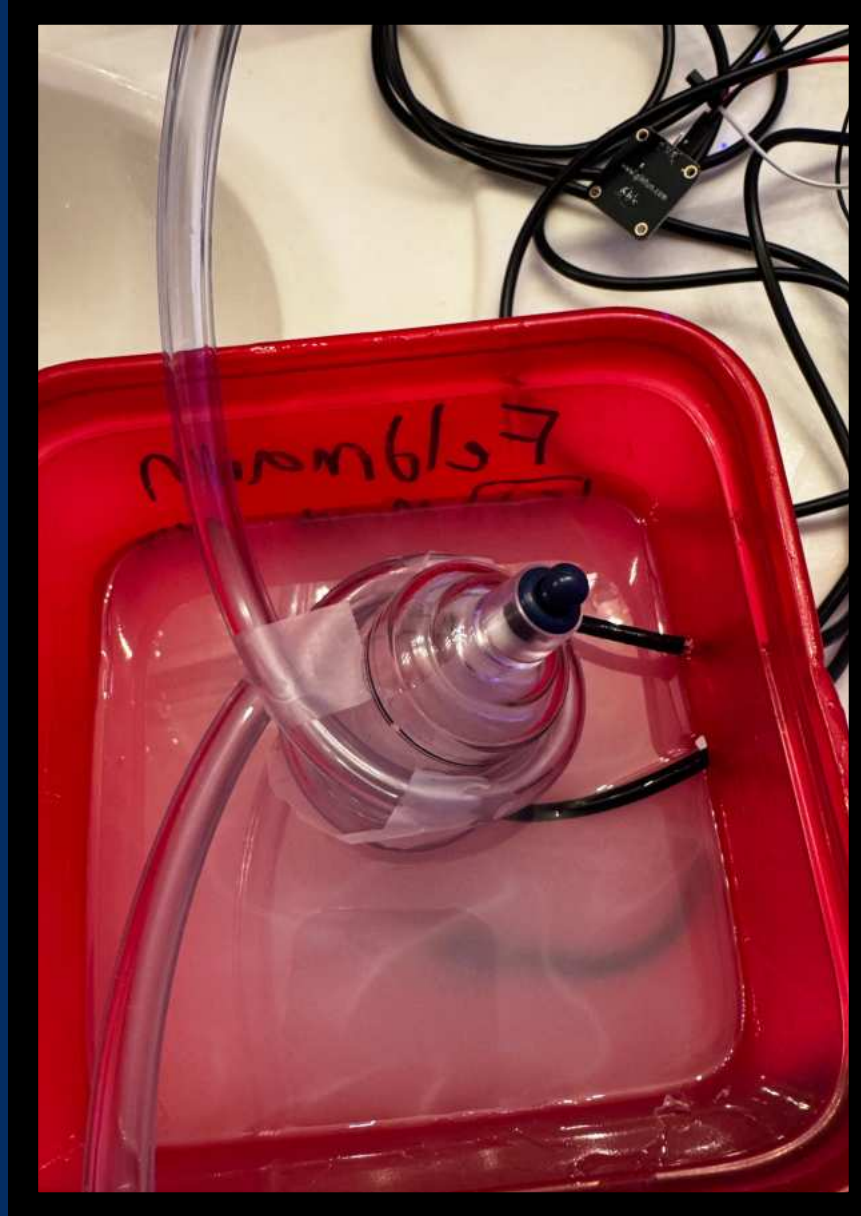


Figure 4. Experimental setup comparing ice therapy and cryogenic cupping.

Traditional ice therapy and cryogenic cupping were applied under controlled conditions to the phantom model. Temperature data was collected over time using an Arduino-based system, with multiple trials conducted to ensure reliable and consistent results across both methods.

Conclusion

Cryogenic cupping reduced surface temperature more than traditional ice and showed slight improvement in mid-layer cooling. However, both methods had minimal effect on deep tissue, indicating that cooling decreases significantly with depth. These results suggest that suction may slightly enhance cooling, but it does not significantly improve deep tissue heat transfer. Overall, cryogenic cupping is more effective at surface-level cooling than deeper tissue cooling.

Extensions



Figure 5. Systems like the Game Ready System use circulating cold fluid and compression to deliver consistent and controlled therapy.

Future improvements could focus on maintaining lower temperatures through continuous fluid circulation, similar to systems like the Game Ready System, which combine controlled cooling with compression. Incorporating a system like this could improve temperature consistency and control. Additionally, adjusting suction strength may help improve cooling depth, while increasing trials and using more realistic tissue models would improve accuracy. This concept could be applied in physical therapy and athletic recovery to develop more effective treatment systems.

References

Bleakley, Chris M., et al. "Should Athletes Return to Sport after Cryotherapy?" British Journal of Sports Medicine, 2012.
Dupuy, Olivier, et al. "Recovery after Exercise: Modalities and Mechanisms." Sports Medicine, 2018.
Khoshnevis, S., et al. "Effects of Cupping Therapy on Pain and Blood Flow." Journal of Traditional Medicine, 2019.

Results

Time (min)	Surface Ice	Surface Cryo	Mid Ice	Mid Cryo	Deep Ice	Deep Cryo
0	93	93	93	93	93	93
2	72	75	82	80	91	91
5	64	66	79	75	90	89
10	63	61	77	73	88	89
15	63	59	77	72	88	87

Figure 5. Data Table of testing between ice pack and cryogenic cups. Cryogenic cupping produced lower temperatures at the surface and mid-layer regions, while deep tissue temperatures changed very little. The largest temperature differences occurred near the surface layer.

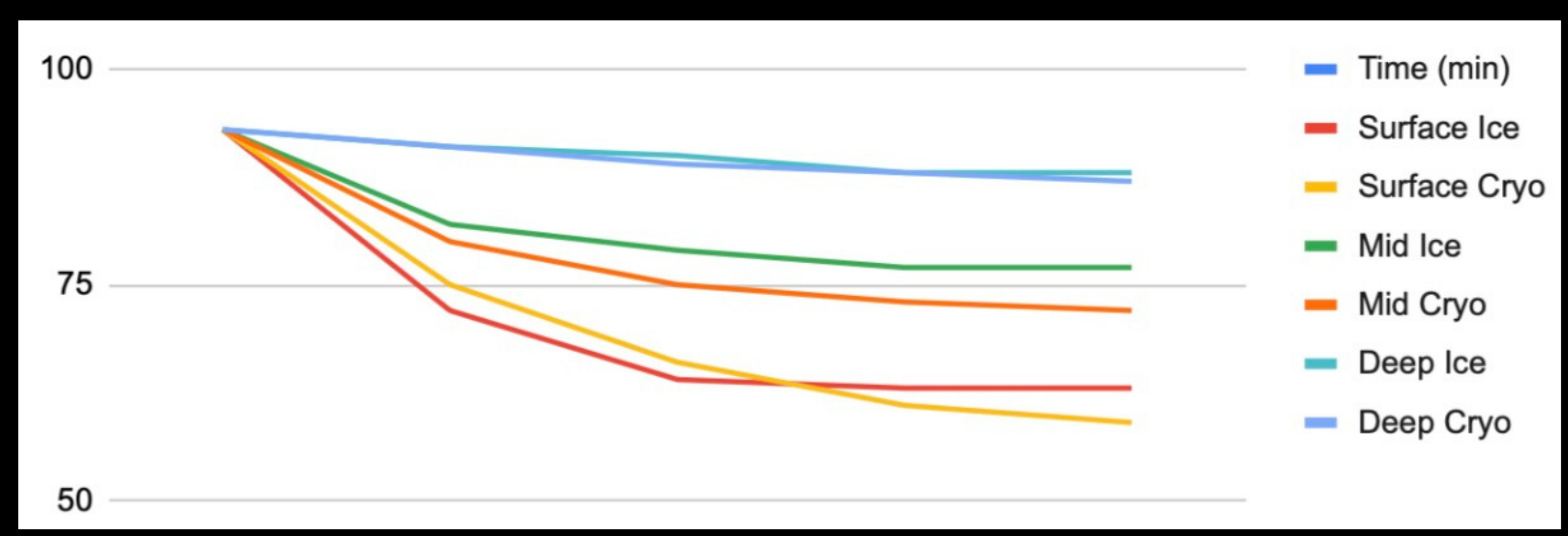


Figure 6. Graph of Comparison between Cryogenic Cups and Ice Pack Temperature. The graph shows that both methods cooled surface tissue quickly, but cryogenic cupping consistently maintained slightly lower temperatures over time. As tissue depth increased, the amount of cooling decreased in both methods, showing limited deep tissue temperature change. Temperature trends remained consistent throughout testing.